

do all the task given in readme that to crct the code and add features as required but after each task give modified files then i will say proceed then do next task

Task 1 is done. Only `game/catch.py` changed.

The bug: `check_catch` only compared `hook.y` and `fish.y` within 10 pixels. It never looked at horizontal position, so a fish on the far side of the screen could be caught.

The fix: The check now uses `hook.get_rect().colliderect(fish.get_rect())`. `Hook` and `Fish` already had `get_rect()`, so no other file needed to change. A catch now needs real overlap in both x and y.

The hook sits at a fixed x (the screen centre), so the fish only get caught as they swim past it.

Say **proceed** when you want Task 2 (multiple fish types).



Catch
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proceed

Task 2 is done. `game/fish.py` and `game/game_engine.py` changed. `catch.py` from Task 1 stays as it was.

Fish types (defined in `FISH_TYPES` in `fish.py`):

Type	Speed	Points	Size	Color
minnow	2 (slow)	10	36×18	blue
bass	3	25	30×16	green
golden	5 (fast)	50	22×12	gold

- **fish.py:** I added a `create_fish(type, x, y, direction)` helper. Each fish now stores its `fish_type`, and `home_y`, the depth it swims at.
- **game_engine.py:** the pond now starts with two of each type at different depths and directions. Scoring already used `hooked_fish.point_value`, so each type awards its own points.
- **Respawn (my addition):** when a fish is scored, a new one of the same type re-enters from the edge it was heading away from, at its original depth. Without this the pond would empty within a few casts, which would hurt the 30-second round in Task 4. It's the small `_respawn` method if you'd rather remove it.

Say **proceed** for Task 3 (player-controlled casting).



Fish

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Game engine

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proceed

Task 3 is done. Three files changed: `game/hook.py`, `game/game_engine.py` and `main.py`.

- **main.py:** pressing **SPACE** now calls `engine.cast()`.
- **game_engine.py:** I removed the automatic re-cast from `update()`, so the hook stays idle until the player acts. The new `cast()` method starts a cast only when the hook is idle. I also added an on-screen "Press SPACE to cast" hint that shows while the hook is idle.
- **hook.py:** `start_cast()` now checks the state itself. It does nothing unless the hook is idle, and it returns `True` or `False` depending on whether a cast started. A second SPACE press mid-cast can't restart or interrupt the hook, even if the engine check were bypassed.

I ran a headless test that confirmed three things:

- The hook stays idle with no input.
- Spamming SPACE mid-cast doesn't interrupt the cast.
- Fish are still caught and scored (score 445 over the test run).

Say **proceed** for Task 4 (30-second round timer).



Hook

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Game engine

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Main

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proceed

Task 4 is done, and that finishes all four tasks. Three files changed: `game/game_engine.py`, `game/renderer.py` and `main.py`.

- **Timer:** each round lasts 30 seconds (`ROUND_SECONDS`). `main.py` passes the real frame time `dt` into `engine.update(dt)`, so the round follows the clock rather than the frame rate. The remaining time shows at the top right as `Time: N`.
- **Round end:** when the time hits zero, `game_over` is set and everything freezes. `cast()` is ignored, and no catches or scoring can happen. A dimmed overlay shows **TIME'S UP!**, the **final score**, and "Press R to play again".
- **Restart:** pressing **R** after the round ends calls `engine.reset()`. That resets the score, the timer, the hook and the fish. I moved the setup out of `__init__` into `reset()` so a new round starts from exactly the same state as the first.
- **Renderer:** I added `draw_overlay()` and a `y_offset` option on `draw_banner()`, so the end screen can show three lines.

I ran a headless test that simulated a full round at 60 FPS: it ended after 1801 frames with `time_left` at 0. The score stayed frozen afterwards, and a reset gave score 0, 30.0 seconds and a working cast. I also checked the end-of-round screen by rendering it to an image.

Design choice: if a fish is still on the line when time runs out, it isn't scored. That matches the README's rule that a fish only counts once it's back at the surface. If you'd rather let the last retract finish, tell me and I'll change it.

Final files (put them in the layout from the README):

- `main.py`
- `game/catch.py`, `game/fish.py`, `game/game_engine.py`, `game/hook.py` and `game/renderer.py` (all in `/mnt/user-data/outputs`)
- `game/__init__.py`, `requirements.txt` and `README.md` are unchanged, so keep your originals.

For the submission, record the "before" video with your original starter files and the "after" video with these. In the "after" clip, show a fish only being caught when it's under the hook, the three fish colours, SPACE being ignored mid-cast, and the countdown ending in the final-score screen.



Game engine
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Renderer
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Main

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